

MEMORY FORENSICS INVESTIGATION REPORT

MemLabs — Lab 5 Challenge

File Name:	MemLab5.docx
Prepared By:	Dnyaneshwari Kale
Date:	30/07/2026
Challenge:	MemLabs Lab 5 — Memory Dump Analysis
Tools Used:	Volatility 3, Volatility 2.6, WinRAR

1. Objective

The objective of this challenge (MemLabs – Lab 5) was to perform memory forensics on a provided memory dump (MemoryDump_Lab5.raw) and recover hidden flag(s) planted inside the image. The investigation involved identifying suspicious processes, extracting files hidden in memory, decoding encoded data, and recovering console/command history to locate the flag(s).

2. Tools Used

The following tools were used during the investigation:

- Volatility 3 (vol.exe) – used first, for initial triage of the memory dump.
- Volatility 2.6 (volatility_2.6_win64_standalone.exe) – used after Volatility 3 could not fully resolve the flag, since it offers legacy plugins (iehistory, consoles, cmdscan) that are not available in Volatility 3.
- WinRAR – used to extract a password-protected/compressed archive recovered from the memory dump.
- Base64 Decoder – used to decode encoded strings recovered from memory into readable flag text.

3. Investigation Log

The investigation was carried out in two stages. In Stage 1, Volatility 3 was used to triage the image, enumerate processes, and attempt to extract and decode the flag. Since Volatility 3 does not implement every legacy Volatility 2 plugin (such as consoles, cmdscan, and iehistory), the analysis was continued in Stage 2 using Volatility 2.6 with the correct OS profile, which allowed the remaining artefacts and the flag to be recovered.

Stage 1 — Analysis with Volatility 3

Step 1: List running processes (pslist)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.pslist
```

Why used: This is the standard starting point of any memory forensics investigation. The pslist plugin walks the kernel's active process list (PsActiveProcessHead) and reports every process that was running at the time the memory image was captured, along with its PID, PPID, thread/handle counts, and creation time. It gives a first overview of what was running on the machine and helps identify unusual or unexpected processes to investigate further.

```

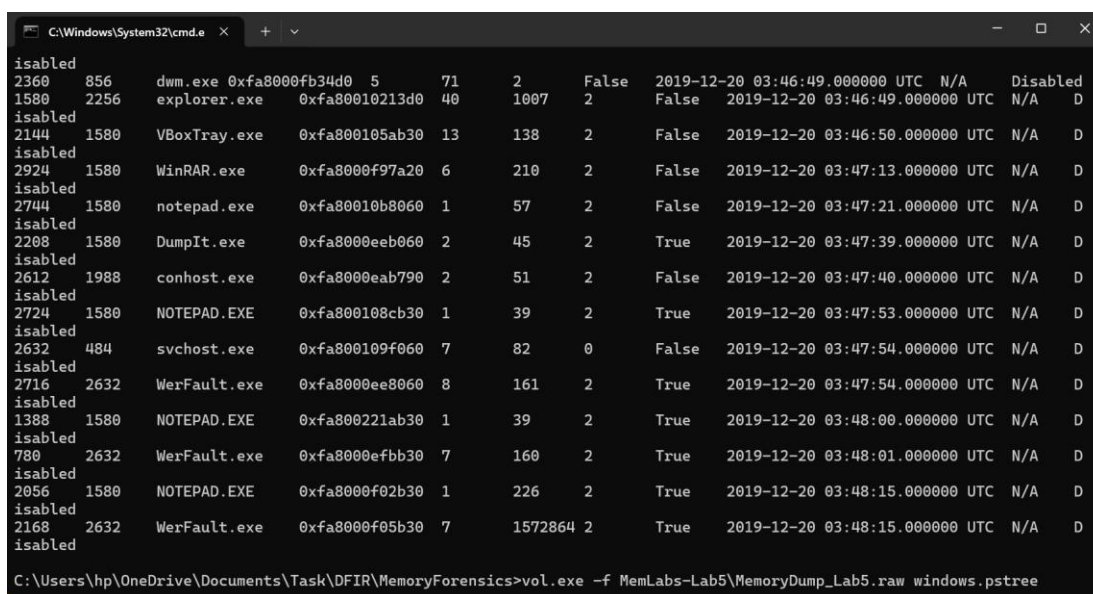
Variable      Value
-----
Kernel Base   0xf80002655000
DTB           0x187000
Symbols file:  C:/Users/hp/OneDrive/Documents/Task/DFIR/MemoryForensics/symbols/windows/ntkrnlmp.pdb/3844DBB920174967BE7AA4A2C20430FA-2.json.xz
Is64Bit       True
IsPAE         False
layer_name    0 WindowsIntel32e
memory_layer  1 FileLayer
KdDebuggerDataBlock 0xf800028460a0
NTBuildLab    7601.17514.amd64fre.win7sp1_rtm.
CSDVersion    1
KdVersionBlock 0xf80002846068
Major/Minor   15.7601
MachineType   34404
KeNumberProcessors 1
SystemTime    2019-12-20 03:47:57+00:00
NtSystemRoot  C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 6
NtMinorVersion 1
PE MajorOperatingSystemVersion 6
PE MinorOperatingSystemVersion 1
PE Machine    34404
PE TimeDateStamp Sat Nov 20 09:30:02 2010

```

Step 2: View process hierarchy (pstree)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.pstree
```

Why used: pstree presents the same process list but arranged as a parent-child tree. Viewing processes hierarchically makes it easier to spot a suspicious process spawned from an unexpected parent, or a legitimate-looking process (e.g. explorer.exe, cmd.exe) that was used to launch something abnormal.



```

C:\Windows\System32\cmd.e  x  +  -  □  x
isabled
2360 856 dmw.exe 0xfa8000fb34d0 5 71 2 False 2019-12-20 03:46:49.000000 UTC N/A Disabled
1580 2256 explorer.exe 0xfa80010213d0 40 1007 2 False 2019-12-20 03:46:49.000000 UTC N/A D
isabled
2144 1580 VBoxTray.exe 0xfa800105ab30 13 138 2 False 2019-12-20 03:46:50.000000 UTC N/A D
isabled
2924 1580 WinRAR.exe 0xfa8000f97a20 6 210 2 False 2019-12-20 03:47:13.000000 UTC N/A D
isabled
2744 1580 notepad.exe 0xfa80010b8060 1 57 2 False 2019-12-20 03:47:21.000000 UTC N/A D
isabled
2208 1580 DumpIt.exe 0xfa8000eeb060 2 45 2 True 2019-12-20 03:47:39.000000 UTC N/A D
isabled
2612 1988 conhost.exe 0xfa8000eab790 2 51 2 False 2019-12-20 03:47:40.000000 UTC N/A D
isabled
2724 1580 NOTEPAD.EXE 0xfa800108cb30 1 39 2 True 2019-12-20 03:47:53.000000 UTC N/A D
isabled
2632 484 svchost.exe 0xfa800109f060 7 82 0 False 2019-12-20 03:47:54.000000 UTC N/A D
isabled
2716 2632 WerFault.exe 0xfa8000ee8060 8 161 2 True 2019-12-20 03:47:54.000000 UTC N/A D
isabled
1388 1580 NOTEPAD.EXE 0xfa800221ab30 1 39 2 True 2019-12-20 03:48:00.000000 UTC N/A D
isabled
780 2632 WerFault.exe 0xfa8000efbb30 7 160 2 True 2019-12-20 03:48:01.000000 UTC N/A D
isabled
2056 1580 NOTEPAD.EXE 0xfa8000f02b30 1 226 2 True 2019-12-20 03:48:15.000000 UTC N/A D
isabled
2168 2632 WerFault.exe 0xfa8000f05b30 7 1572864 2 True 2019-12-20 03:48:15.000000 UTC N/A D
isabled
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.pstree

```

Step 3: Check command-line arguments (cmdline)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.cmdline
```

Why used: The cmdline plugin recovers the exact command-line string each process was launched with. This is important in CTF-style memory forensics because attackers/challenge authors often hide clues, file paths, or encoded strings directly in the arguments used to start a process (for example, opening a specific file in Notepad or running a script with a suspicious parameter).

```
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.pstree
Volatility 3 Framework 2.28.0
Progress: 100.00
PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime A
udit Cmd Path
4 0 System 0xfa8000ca0040 80 562 N/A False 2019-12-20 03:41:40.000000 UTC N/A - -
* 248 4 smss.exe 0xfa80014b9040 3 37 N/A False 2019-12-20 03:41:40.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\smss.exe \SystemRoot\System32\smss.exe \SystemRoot\System32\smss.exe
** 376 248 psxss.exe 0xfa8001c6b060 18 786 0 False 2019-12-20 03:41:47.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\psxss.exe %SystemRoot%\system32\psxss.exe C:\Windows\system32\psxss.exe
320 312 csrss.exe 0xfa8000d96b30 10 471 0 False 2019-12-20 03:41:45.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedS
ection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,
3 ServerDll=winsrv:ConServerDllInitialization,2 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16 C:\Windo
ws\system32\csrss.exe
368 360 csrss.exe 0xfa8001c5d060 8 173 1 False 2019-12-20 03:41:47.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedS
ection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,
3 ServerDll=winsrv:ConServerDllInitialization,2 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16 C:\Windo
ws\system32\csrss.exe
416 360 winlogon.exe 0xfa8001c775a0 3 108 1 False 2019-12-20 03:41:48.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\winlogon.exe winlogon.exe C:\Windows\system32\winlogon.exe
428 312 wininit.exe 0xfa8001c7c060 3 75 0 False 2019-12-20 03:41:48.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\wininit.exe wininit.exe C:\Windows\system32\wininit.exe
* 500 428 lsm.exe 0xfa8001cbc4a0 10 171 0 False 2019-12-20 03:41:50.000000 UTC N/A \Device\
HarddiskVolume2\Windows\System32\lsm.exe C:\Windows\system32\lsm.exe C:\Windows\system32\lsm.exe
* 484 428 services.exe 0xfa8001c7bb30 8 213 0 False 2019-12-20 03:41:50.000000 UTC N/A \
```

```
C:\Windows\System32\cmd.e x + v
ection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,
3 ServerDll=winsrv:ConServerDllInitialization,2 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16 C:\Windo
ws\system32\csrss.exe
* 2612 1988 conhost.exe 0xfa8000eab790 2 51 2 False 2019-12-20 03:47:40.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\conhost.exe \??\C:\Windows\system32\conhost.exe C:\Windows\system32\conh
ost.exe
2120 364 winlogon.exe 0xfa8000f82060 4 113 2 False 2019-12-20 03:46:42.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\winlogon.exe winlogon.exe C:\Windows\system32\winlogon.exe
1580 2256 explorer.exe 0xfa80010213d0 40 1007 2 False 2019-12-20 03:46:49.000000 UTC N/A \
Device\HarddiskVolume2\Windows\explorer.exe C:\Windows\Explorer.EXE C:\Windows\Explorer.EXE
* 2144 1580 VBoxTray.exe 0xfa800105ab30 13 138 2 False 2019-12-20 03:46:50.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\VBoxTray.exe "C:\Windows\System32\VBoxTray.exe" C:\Windows\System32\VBox
Tray.exe
* 2208 1580 DumpIt.exe 0xfa8000eeb060 2 45 2 True 2019-12-20 03:47:39.000000 UTC N/A \
Device\HarddiskVolume2\Users\SmartNet\Downloads\DumpIt\DumpIt.exe "C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe"
C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe
* 2724 1580 NOTEPAD.EXE 0xfa800108cb30 1 39 2 True 2019-12-20 03:47:53.000000 UTC N/A \
Device\HarddiskVolume2\Users\SmartNet\Videos\NOTEPAD.EXE "C:\Users\SmartNet\Videos\NOTEPAD.EXE" C:\Users\SmartNe
t\Videos\NOTEPAD.EXE
* 2056 1580 NOTEPAD.EXE 0xfa8000f02b30 1 226 2 True 2019-12-20 03:48:15.000000 UTC N/A \
Device\HarddiskVolume2\Users\SmartNet\Videos\NOTEPAD.EXE - -
* 2924 1580 WinRAR.exe 0xfa8000f97a20 6 210 2 False 2019-12-20 03:47:13.000000 UTC N/A \
Device\HarddiskVolume2\Program Files\WinRAR\WinRAR.exe "C:\Program Files\WinRAR\WinRAR.exe" "C:\Users\SmartNet\Document
s\SW1wb3J0YW50.rar" C:\Program Files\WinRAR\WinRAR.exe
* 1388 1580 NOTEPAD.EXE 0xfa800221ab30 1 39 2 True 2019-12-20 03:48:00.000000 UTC N/A \
Device\HarddiskVolume2\Users\SmartNet\Videos\NOTEPAD.EXE - -
* 2744 1580 notepad.exe 0xfa80010b8060 1 57 2 False 2019-12-20 03:47:21.000000 UTC N/A \
Device\HarddiskVolume2\Windows\System32\notepad.exe "C:\Windows\system32\notepad.exe" C:\Windows\system32\note
pad.exe
```

Step 4: Search for a Notepad handle (filescan + findstr)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.filescan | findstr /i
"NOTEPAD.EXE"
```

Why used: filescan enumerates every FILE_OBJECT structure found in memory, effectively listing files that were open (even if since closed on disk) at the time of capture. Piping it through findstr /i "NOTEPAD.EXE" narrows this huge list down to only entries related to Notepad, which is a strong indicator that a text file was opened/edited by the user — a common way challenge authors leave a note or a clue in memory.

```

C:\Windows\System32\cmd.exe
2296 svchost.exe C:\Windows\System32\svchost.exe -k LocalServicePeerNet
2572 WmiPrvSE.exe C:\Windows\system32\wbem\wmiPrvse.exe
2940 sppsvc.exe C:\Windows\system32\sppsvc.exe
1128 svchost.exe C:\Windows\System32\svchost.exe -k secsvcs
668 dllhost.exe C:\Windows\system32\DllHost.exe /Processid:{76D0CB12-7604-4048-B83C-1005C7DDC503}
2608 SearchFilterHost.exe "C:\Windows\system32\SearchFilterHost.exe" 0 504 508 516 65536 512
628 SearchProtocolHost.exe "C:\Windows\system32\SearchProtocolHost.exe" Global\UsGthrFltPipeMssGthrPipe3_Global\UsGthrCtrl
FltPipeMssGthrPipe3 1 -2147483646 "Software\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT; MS
Search 4.0 Robot)" "C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "DownLevelDaemon"
1988 csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On
SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winhttpd\WinHttpd.dll ServerDll=winhttpd\WinHttpd.dll
initialization,2 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
2120 winlogon.exe winlogon.exe
1968 taskhost.exe "taskhost.exe"
2360 dwm.exe "C:\Windows\system32\Dwm.exe"
1580 explorer.exe C:\Windows\Explorer.EXE
2144 VBoxTray.exe "C:\Windows\System32\VBoxTray.exe"
2924 WinRAR.exe "C:\Program Files\WinRAR\WinRAR.exe" "C:\Users\SmartNet\Documents\SW1wb3J0YW50.rar"
2744 notepad.exe "C:\Windows\system32\notepad.exe"
2208 DumpIt.exe "C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe"
2612 conhost.exe "C:\Windows\system32\conhost.exe"
2724 NOTEPAD.EXE "C:\Users\SmartNet\Videos\NOTEPAD.EXE"
2632 svchost.exe C:\Windows\System32\svchost.exe -k WerSvcGroup
2716 WerFault.exe C:\Windows\SysWOW64\WerFault.exe -u -p 2724 -s 156
1388 NOTEPAD.EXE -
780 WerFault.exe C:\Windows\SysWOW64\WerFault.exe -u -p 1388 -s 156
2056 NOTEPAD.EXE -
2168 WerFault.exe -
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>

```

Step 5: Dump the suspected file by virtual address (dumpfiles --virtaddr)

```
mkdir output vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw -o output windows.dumpfiles
--virtaddr 0x3fca5250
```

Why used: An output folder is first created to hold extracted artefacts. windows.dumpfiles --virtaddr then extracts the actual file content associated with the FILE_OBJECT located at the specified virtual address (obtained from the filescan output above), writing it to disk so it can be opened and inspected outside of Volatility.

```

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.filescan | findstr /i "N
OTEPAD.EXE"
0x3ee9d070 100.0\Users\SmartNet\Videos\NOTEPAD.EXEHed
0x3faaf440 \Windows\System32\en-US\notepad.exe.mui
0x3fb8b440 \Windows\System32\en-US\notepad.exe.mui
0x3fca5250 \Users\SmartNet\Videos\NOTEPAD.EXE
0x3fda1d00 \Windows\System32\notepad.exe

```

Step 6: Dump a file by name filter (dumpfiles --filter)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw -o output windows.dumpfiles --filter
"SW1wb3J0YW50"
```

Why used: "SW1wb3J0YW50" is Base64 for the word "Important" — a filename spotted during file scanning. The --filter option lets dumpfiles target and extract only the file object(s) whose name matches this string, pulling out a file named Important (later identified as Important.rar) directly from the memory image for offline analysis.

```

Directory of C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics
07/30/2026 11:34 PM <DIR>      .
07/30/2026 11:22 PM <DIR>      ..
07/22/2026 03:55 PM          1,306,258 all_vads.txt
07/22/2026 03:14 PM          467,256 dlllist.txt
07/22/2026 03:20 PM <DIR>      dump
07/22/2026 12:02 PM          3,956 LICENSE.txt
07/30/2026 10:23 PM <DIR>      MemLabs-Lab5
07/30/2026 10:22 PM          185,946,989 MemLabs-Lab5.7z
07/30/2026 11:34 PM <DIR>      output
07/22/2026 03:28 PM          0 PE.0xfa8005a1d9e0.3496.0x13e397190.dmp
07/22/2026 03:35 PM          90,112 PE.0xfa8005a1d9e0.3496.0x400000.dmp
07/22/2026 12:02 PM          5,888 README.md
07/22/2026 12:07 PM <DIR>      symbols
07/22/2026 11:43 AM          5,368,709,120 Triage-Memory.mem
07/22/2026 03:45 PM          14,216 vadinfo.txt
07/22/2026 12:02 PM          22,216,348 vol.exe
07/22/2026 12:02 PM          22,292,045 volshell.exe
        11 File(s)  5,601,052,188 bytes
        6 Dir(s)  33,506,754,560 bytes free

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw -o output windows.dmpfiles --fi
rtaddr 0x3fca5250
Volatility 3 Framework 2.28.0
Progress: 100.00          PDB scanning finished
Cache  FileObject      FileName      Result

```

Step 7: Inspect process memory regions (vadinfo)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.vadinfo --pid 2724
```

Why used: vadinfo lists the Virtual Address Descriptors (VADs) of a chosen process — i.e. the memory regions it has mapped, along with their protection flags (e.g. PAGE_EXECUTE_READWRITE) and backing files. Running it against PID 2724 was used to check whether that process had any suspicious, injected, or unusually-permissioned memory regions worth extracting.

```

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw -o output windows.dmpfiles --fi
lter "SM1wb3J0YW50"
Volatility 3 Framework 2.28.0
Progress: 100.00          PDB scanning finished
Cache  FileObject      FileName      Result

```

Step 8: List loaded DLLs (dlllist)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.dlllist --pid 2724
```

Why used: dlllist enumerates every DLL module loaded into the address space of PID 2724. This was used to verify whether the process had loaded any unexpected or non-standard modules, which can indicate code injection or the use of an unusual library relevant to the challenge.

```

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.vadinfo --pid 2724
Volatility 3 Framework 2.28.0
Progress: 100.00          PDB scanning finished
Cache  FileObject      FileName      Result

```

PID	Process	Offset	Start VPN	End VPN	Tag	Protection	CommitCharge	PrivateMemory	Parent	File	File output
2724	NOTEPAD.EXE	0xfffffa8001d06170		0x75220000		0x75265fff	Vad	PAGE_EXECUTE_WRITECOPY	3	0	0xfa800108cf78
2724	NOTEPAD.EXE	0xfffffa80020361f0		0x1000000		0x1013fff	Vad	PAGE_EXECUTE_WRITECOPY	3	0	0xfa80010d06170
2724	NOTEPAD.EXE	0xfffffa8000ee40b0		0xd0000	0xd3fff	Vad	PAGE_READONLY	0	0	0xfa80020361f0	N/A
2724	NOTEPAD.EXE	0xfffffa8000ee2680		0x40000	0x40fff	Vad	PAGE_EXECUTE_WRITECOPY	0	0	0xfa8000ee40b0	\Wind
2724	NOTEPAD.EXE	0xfffffa80021cd690		0x20000	0x20fff	Vadm	PAGE_READWRITE	1	1	0xfa8000ee2680	N/A
2724	NOTEPAD.EXE	0xfffffa8002289470		0x10000	0x1ffff	Vad	PAGE_READWRITE	0	0	0xfa80021cd690	N/A
2724	NOTEPAD.EXE	0xfffffa8002221a80		0x30000	0x30fff	Vadm	PAGE_READWRITE	1	1	0xfa80021cd690	N/A
2724	NOTEPAD.EXE	0xfffffa8001019290		0x50000	0x8ffff	VadS	PAGE_READWRITE	7	1	0xfa8000ee2680	N/A

Step 9: Scan for network activity (netscan)

```
vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.netscan
```

Why used: netscan searches the memory image for network-related pool structures (TCP endpoints, listeners, sockets) so that any network connections active at capture time can be reviewed. This step was used to rule out network exfiltration or a remote connection being relevant to the flag.

```
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.dlllist --pid 2724
Volatility 3 Framework 2.28.0
Progress: 100.00
PDB scanning finished
PID Process Base Size PDB Name Path LoadCount LoadTime File output
2724 NOTEPAD.EXE 0x1000000 0x140000 NOTEPAD.EXE C:\Users\SmartNet\Videos\NOTEPAD.EXE -1 N/A Disabled
2724 NOTEPAD.EXE 0x76da0000 0x1a9000 ntdll.dll C:\Windows\SYSTEM32\ntdll.dll -1 N/A Disabled
2724 NOTEPAD.EXE 0x74790000 0x3f000 wow64.dll C:\Windows\SYSTEM32\wow64.dll 3 2019-12-20 03:47:53.000000 UT
C Disabled
2724 NOTEPAD.EXE 0x73e20000 0x5c000 wow64win.dll C:\Windows\SYSTEM32\wow64win.dll 1 2019-12-20 03:47:53.0
00000 UTC Disabled
2724 NOTEPAD.EXE 0x74800000 0x8000 wow64cpu.dll C:\Windows\SYSTEM32\wow64cpu.dll 1 2019-12-20 03:47:53.0
00000 UTC Disabled
2724 NOTEPAD.EXE 0x1000000 0x140000 NOTEPAD.EXE C:\Users\SmartNet\Videos\NOTEPAD.EXE - - Disabled
2724 NOTEPAD.EXE 0x76f80000 0x180000 ntdll.dll C:\Windows\SysWOW64\ntdll.dll - - Disabled
2724 NOTEPAD.EXE 0x75690000 0x110000 kernel32.dll C:\Windows\syswow64\kernel32.dll - - Disab
led
2724 NOTEPAD.EXE 0x75220000 0x46000 KERNELBASE.dll C:\Windows\syswow64\KERNELBASE.dll - - Disabled
2724 NOTEPAD.EXE 0x75040000 0x7b000 cmdlg32.dll C:\Windows\syswow64\cmdlg32.dll - - Disabled
2724 NOTEPAD.EXE 0x752d0000 0xac000 msvcrt.dll C:\Windows\syswow64\msvcrt.dll - - Disabled
2724 NOTEPAD.EXE 0x751c0000 0x57000 SHLWAPI.dll C:\Windows\syswow64\SHLWAPI.dll - - Disabled
2724 NOTEPAD.EXE 0x74d20000 0x90000 GDI32.dll C:\Windows\syswow64\GDI32.dll - - Disabled
2724 NOTEPAD.EXE 0x76840000 0x100000 USER32.dll C:\Windows\syswow64\USER32.dll - - Disabled
2724 NOTEPAD.EXE 0x76ae0000 0xa0000 ADVAPI32.dll C:\Windows\syswow64\ADVAPI32.dll - - N/A Disabled
2724 NOTEPAD.EXE 0x75380000 0x19000 sechost.dll C:\Windows\SysWOW64\sechost.dll - - Disabled
2724 NOTEPAD.EXE 0x74db0000 0xf0000 RPCRT4.dll C:\Windows\syswow64\RPCRT4.dll - - Disabled
2724 NOTEPAD.EXE 0x74ae0000 0x60000 SspiCli.dll C:\Windows\syswow64\SspiCli.dll - - Disabled
2724 NOTEPAD.EXE 0x74ad0000 0xc0000 CRYPTBASE.dll C:\Windows\syswow64\CRYPTBASE.dll - - Disabled
2724 NOTEPAD.EXE 0x74bb0000 0xa000 LPK.dll C:\Windows\syswow64\LPK.dll - - Disabled
2724 NOTEPAD.EXE 0x74c80000 0x9d000 USP10.dll C:\Windows\syswow64\USP10.dll - - Disabled
2724 NOTEPAD.EXE 0x73c80000 0x19e000 COMCTL32.dll C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b
64144ccf1df_6.0.7601.17514_none_41e6975e2bd6f2b2\COMCTL32.dll - - Disabled
```

Step 10: Search raw memory for an encoded flag fragment (findstr)

```
findstr /i "Zmxh" MemLabs-Lab5\MemoryDump_Lab5.raw
```

Why used: "Zmxh" is the Base64 encoding of the letters "fla" — the start of the word "flag". Instead of relying only on Volatility plugins, the raw memory dump was searched directly with findstr for this Base64 fragment, on the chance that a Base64-encoded flag string was sitting in memory outside of a normal file structure.

```
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics>vol.exe -f MemLabs-Lab5\MemoryDump_Lab5.raw windows.netscan
Volatility 3 Framework 2.28.0
Progress: 100.00
PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
0x3e82b3c0 UDPv6 ::1 55322 * 0 1372 svchost.exe 2019-12-20 03:43:51.000000 UTC
0x3e82d3d0 UDPv4 127.0.0.1 55324 * 0 1372 svchost.exe 2019-12-20 03:43:51.000000 UTC
0x3e83ed70 UDPv4 0.0.0.0 55325 * 0 340 svchost.exe 2019-12-20 03:44:11.000000 UTC
0x3e85f010 TCPv4 127.0.0.1 2869 127.0.0.1 49162 CLOSED 4 System N/A
0x3e8888c0 UDPv4 0.0.0.0 55326 * 0 340 svchost.exe 2019-12-20 03:44:11.000000 UTC
0x3e8888c0 UDPv6 ::1 55326 * 0 340 svchost.exe 2019-12-20 03:44:11.000000 UTC
0x3ea02320 TCPv4 0.0.0.0 9 0.0.0.0 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea02320 TCPv6 ::1 9 ::1 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea02ad0 TCPv4 0.0.0.0 9 0.0.0.0 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea09440 UDPv4 0.0.0.0 9 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea09440 UDPv6 ::1 9 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea09bb0 UDPv4 0.0.0.0 9 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea0ead0 TCPv4 0.0.0.0 13 0.0.0.0 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea172c0 UDPv4 0.0.0.0 19 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea17720 TCPv4 0.0.0.0 19 0.0.0.0 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea17720 TCPv6 ::1 19 ::1 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea1c2c0 TCPv4 0.0.0.0 17 0.0.0.0 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea1cbb0 UDPv4 0.0.0.0 19 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea1cbb0 UDPv6 ::1 19 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea22ec0 UDPv4 10.0.2.15 137 * 0 4 System 2019-12-20 03:42:17.000000 UTC
0x3ea23170 UDPv4 0.0.0.0 17 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea23170 UDPv6 ::1 17 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea23670 UDPv4 0.0.0.0 17 * 0 1416 TCPSVCS.EXE 2019-12-20 03:42:14.000000 UTC
0x3ea23ad0 TCPv4 0.0.0.0 17 0.0.0.0 0 LISTENING 1416 TCPSVCS.EXE -
0x3ea23ad0 TCPv6 ::1 17 ::1 0 LISTENING 1416 TCPSVCS.EXE -
```

Note: At this point Volatility 3 had provided several leads (the Important.rar file, process 2724's memory, etc.) but not a directly usable flag. Volatility 2 was therefore brought in, since it supports additional legacy plugins (imageinfo, consoles, cmdscan, iehistory) that Volatility 3 does not, and these were needed to fully trace the flag.

Stage 2 — Analysis with Volatility 2.6

Step 11: Identify the memory profile (imageinfo)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw imageinfo
```

Why used: Unlike Volatility 3 (which auto-detects the OS), Volatility 2 requires the analyst to manually supply a --profile matching the operating system of the captured machine. imageinfo inspects the memory image's structures (KDBG, DTB, etc.) and suggests the most likely matching profile(s) — here, Win7SP1x64 — which is required for every subsequent Volatility 2 command to parse the memory correctly.

```

yarascan Scan process or kernel memory with Yara signatures

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>volatility 2.6_win64_standalone.exe -f MemoryDump_Lab5.raw imageinfo
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP1x64_23418
AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
AS Layer2 : FileAddressSpace (C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5\MemoryDump_Lab5.raw)
PAE type : No PAE
DTB : 0x187000L
KDBG : 0xf800028460a0L
Number of Processors : 1
Image Type (Service Pack) : 1
KPCR for CPU 0 : 0xfffff80002847d00L
KUSER_SHARED_DATA : 0xfffff80000000000L
Image date and time : 2010-12-20 03:47:57 UTC+0000
Image local date and time : 2019-12-20 09:17:57 +0530

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>volatility 2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=
Volatility Foundation Volatility Framework 2.6
Offset(V) Name PID PPID Thds Hnds Sess Wow64 Start Exit

```

Step 12: Re-confirm the process list with the correct profile (pslist)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 pslist
```

Why used: With the correct profile now known, pslist was re-run under Volatility 2 to confirm the process list is parsed correctly and to cross-check it against the Volatility 3 output from Stage 1.

C:\Windows\System32\cmd.exe																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
-----------------------------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Step 13: Recover Internet Explorer history (iehistory)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64  
iehistory
```

Why used: iehistory is a Volatility 2 plugin (not available in Volatility 3) that reconstructs Internet Explorer's cache/history records from process memory. It was used to check whether the user had browsed to or downloaded anything relevant to the challenge through the browser.

```
C:\Users\hnp\OneDrive\Documents\Task\NETRA\MemoryForensics\MemLabs-Lab5>volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 iehistory
Volatility Foundation Volatility Framework 2.6
*****
Process: 1396 explorer.exe
Cache type "DESI" at b635910cf
Last modified: 2019-12-28 09:16:37 UTC+0000
Last accessed: 2019-12-28 03:46:38 UTC+0000
URL: Alissa Simpson@file:///C:/Windows/AppPatch/ZmxhZ3shIV93M6vMX2QwbJNfU3Q0ZzhtNV8tZl9MNEJfNFVhYXNlYXQ4YyEhFQ.bmp
C:\Users\hnp\OneDrive\Documents\Task\NETRA\MemoryForensics\MemLabs-Lab5>
```

Step 14: Search for AppPatch-related files (filescan + findstr)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64  
filesCAN | findstr /i "AppPatch"
```

Why used: The Volatility 2 filescan plugin was used again (this time under the correct profile) and filtered for the string "AppPatch", a legitimate Windows system directory that is sometimes reused by challenge authors to hide a file in plain sight among genuine system paths.

[illegible]

Step 15: Read the dumped file's contents (type)

```
type output\file.None.0xffffffffa800209bc40.dat
```

Why used: After dumping the file object found above (address 0xfffffa800209bc40), the Windows type command was used to print its raw contents directly to the console. This is a quick way to read a small extracted file without needing a separate viewer.

```
C:\Windows\System32\cmd.e
<keyCreationFlags>%keyCreationFlags%
<keyPromptFlags>%keyPromptFlags%
</UI>
<Security>
  <lockbacklocking>
    <lockAfterTime>%lockAfterTime%
    <lockAfterGlobalTime>%lockAfterGlobalTime%
  </lockbacklocking>
  <Policy />
  <MasterPassword>
    <MinimumLength>%MinimumLength%
    <MinimumQuality>%MinimumQuality%
  </MasterPassword>
</Security>
<Native />
<PasswordGenerators>
  <autoGeneratePasswordsProfile>
    <GeneratorType>CharSet/GeneratorType>
    <Length>20/<Length>
    <CharSetRanges>ILD_____/CharSetRanges>
  </autoGeneratePasswordsProfile>
  <LastUsedProfile>
    <GeneratorType>CharSet/GeneratorType>
    <Length>20/<Length>
    <CharSetRanges>ILD_____/CharSetRanges>
  </LastUsedProfile>
  <UserProfiles />
</PasswordGenerators>
<Default>
  <OptionsTabIndex>%OptionsTabIndex>
  <SearchParameters>
    <ComparisonMode>InvariantCultureIgnoreCase/ComparisonMode>
  </SearchParameters>
  <KeySources>
    <Association>
      <DatabasePath>.\..\\Users\\SmartNet\\Secrets\\Hidden.kdbx</DatabasePath>
      <Password>true/Password>
    </Association>
  </KeySources>
</Default>
<Integration>
  <BuiltInSchemasOverrides>
    <BuiltInOverridesEnabled>1/BuiltInOverridesEnabled>
    <CustomOverrides />
  </BuiltInSchemasOverrides>

```

Step 16: Dump the suspicious process's memory (memdump)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 memdump -p 1396 -D output
```

Why used: memdump extracts the complete memory space (all pages) of a chosen process — here PID 1396 — into a single .dmp file in the output folder. This gives a full offline copy of everything that process held in memory, which can then be searched for strings/text that never touched disk as a separate file.

[illegible]

Decoding the Recovered Flag (Stage 1 complete)

A Base64-encoded string was located inside the dumped data:

ZmxhZ3shIV93M0xMX20wbjNfU300ZzMtMV8wZl19MNEJfNV9EMG4zXyEhf0

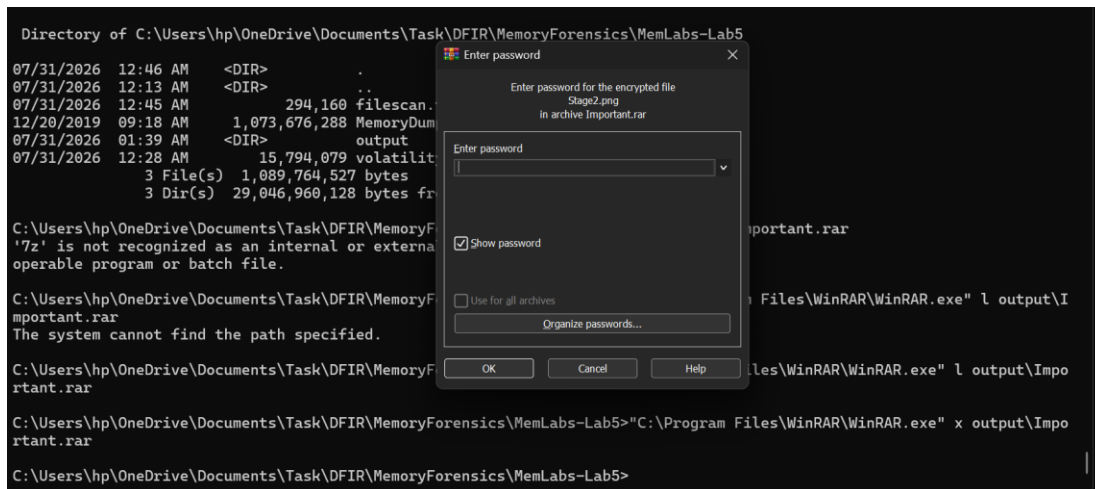
Decoding this Base64 string revealed the first flag of the challenge:

```
f1ag{!! w3LL d0n3 St4g3-1 0f L4B 5 D0n3 !!}
```

Step 17: Extract the recovered RAR archive (WinRAR)

```
"C:\Program Files\WinRAR\WinRAR.exe" x output\Important.rar
```

Why used: The Important.rar file that had been carved out of memory earlier (Step 6) was extracted using WinRAR's command-line extraction mode (x), to access the file(s) it contained and continue the investigation toward the next stage of the challenge.



Step 18: Recover typed console commands (consoles)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64  
consoles
```

Why used: consoles is a Volatility 2 plugin that reconstructs the screen buffer and command history of any cmd.exe console sessions that were open in memory. It was used to see exactly what the user had typed into

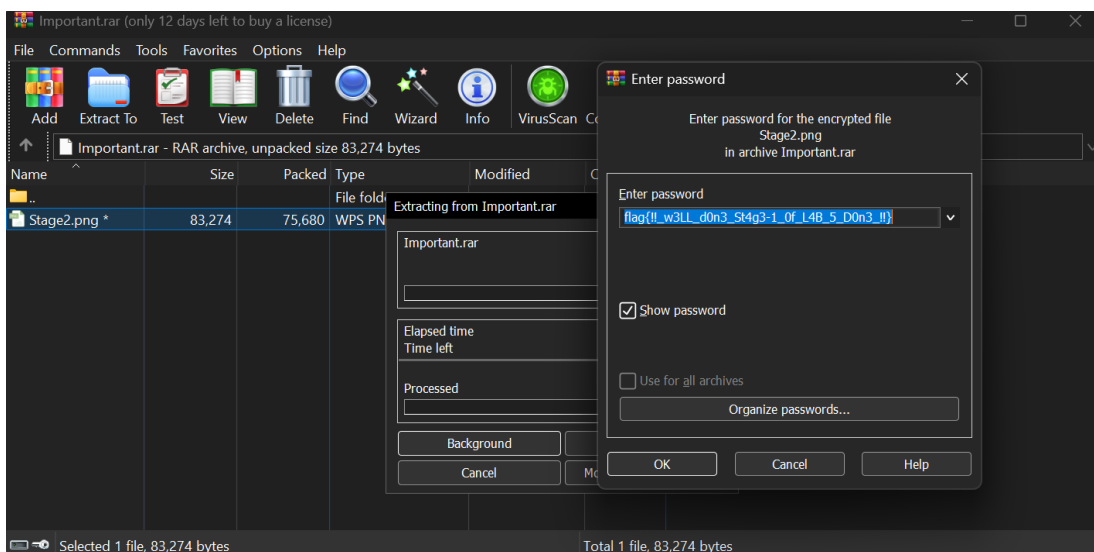
the command prompt, which can reveal manually entered commands, paths, or additional flag text that would not appear in any file on disk.

```
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 consoles
Volatility Foundation Volatility Framework 2.6
*****
ConsoleProcess: conhost.exe Pid: 2612
Console: 0xfffff6200 CommandHistorySize: 50
HistoryBufferCount: 1 HistoryBufferMax: 4
OriginalTitle: C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe
Title: C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe
AttachedProcess: DumpIt.exe Pid: 2208 Handle: 0x60
-----
CommandHistory: 0x18ea90 Application: DumpIt.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x60
-----
Screen 0x171050 X:80 Y:300
Dump:
DumpIt - v1.3.2.20110401 - One click memory memory dumper
Copyright (c) 2007 - 2011, Matthieu Suiche <http://www.msuiche.net>
Copyright (c) 2010 - 2011, MoonSols <http://www.moonsols.com>
```

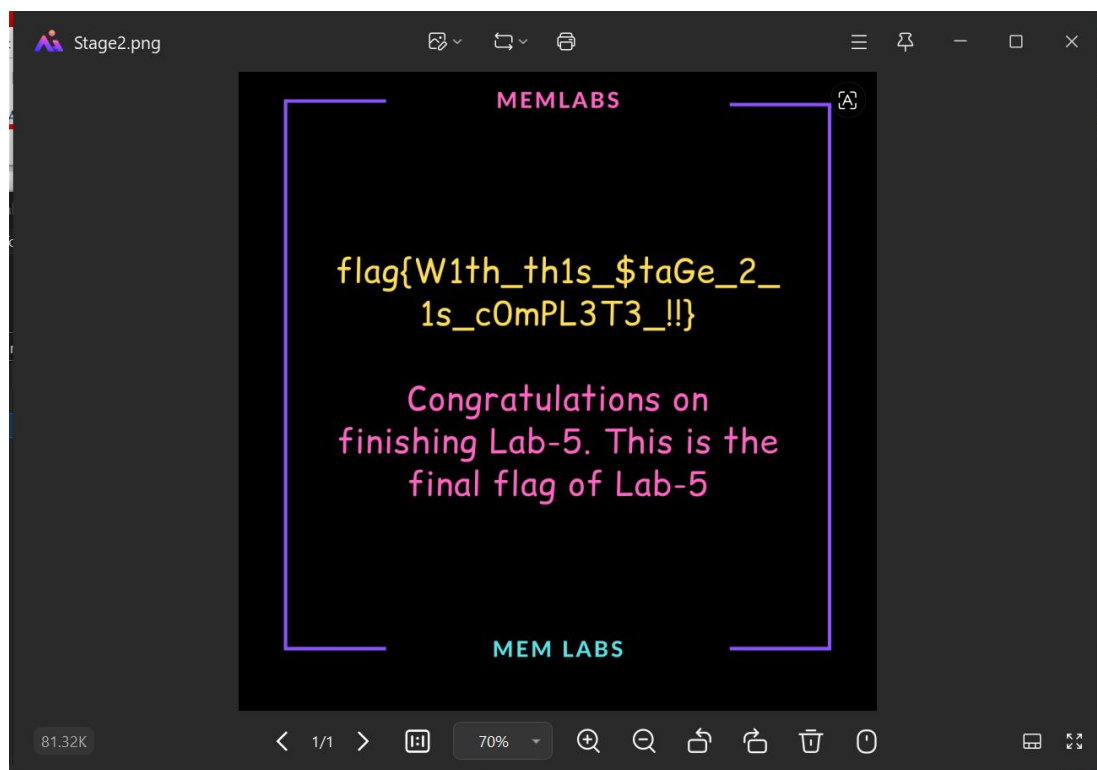
Step 19: Cross-check with command history (cmdscan)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 cmdscan
```

Why used: cmdscan scans memory for `COMMAND_HISTORY` structures used by the Windows console host, providing an alternative recovery path for previously typed commands. It was run alongside consoles to cross-verify the recovered command history that led to the RAR archive above.



The password recovered from the console history was used to open the RAR archive, revealing Stage2.png, which contained the Stage 2 flag:



flag{W1th_th1s_\$taGe_2_1s_cOmPL3T3_!!}

Stage 3 — Recovering the PE image and static reverse engineering

The Stage 2 image indicated that the challenge was not yet fully solved. Unlike the earlier Volatility 3 dumpfiles step (Step 6), which pulled the plain data content of a file out of memory for reading, this step targets the same NOTEPAD.EXE object again but specifically extracts its ImageSectionObject — the mapped, in-memory copy of the executable's own PE image — so it can be reconstructed as a runnable/loadable .exe and reverse-engineered. This is a different artefact and a different goal from Step 6, even though both start from the same file object.

Step 20: Dump the NOTEPAD.EXE image section to a folder (dumpfiles -Q -D)

```
volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64  
dumpfiles -Q 0x3fca5250 -D notepad-dump
```

Why used: The Volatility 2 dumpfiles plugin's -Q flag dumps every memory-resident section (ImageSectionObject, DataSectionObject, SharedCacheMap) associated with the file object at the given physical/virtual address, and -D specifies the destination folder. The first attempt failed because the notepad-dump folder did not exist yet ("notepad-dump is not a directory"); after creating it with mkdir, the command was re-run successfully, producing three files, including a .img file — the reconstructed image section of NOTEPAD.EXE.

```

C:\Windows\System32\cmd.exe
C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 dumpfiles -Q 0x3fca5250 -D notepad-dump
Volatility Foundation Volatility Framework 2.6
ERROR : volatility.debug : notepad-dump is not a directory

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>mkdir notepad-dump

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>volatility_2.6_win64_standalone.exe -f MemoryDump_Lab5.raw --profile=Win7SP1x64 dumpfiles -Q 0x3fca5250 -D notepad-dump
Volatility Foundation Volatility Framework 2.6
ImageSectionObject 0x3fca5250 None \Device\HarddiskVolume2\Users\SmartNet\Videos\NOTEPAD.EXE
DataSectionObject 0x3fca5250 None \Device\HarddiskVolume2\Users\SmartNet\Videos\NOTEPAD.EXE
SharedCacheMap 0x3fca5250 None \Device\HarddiskVolume2\Users\SmartNet\Videos\NOTEPAD.EXE

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>dir notepad-dump
Volume in drive C has no label.
Volume Serial Number is 18CC-2E04

Directory of C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5\notepad-dump

07/31/2026 10:12 AM <DIR> .
07/31/2026 10:12 AM <DIR> ..
07/31/2026 10:12 AM 69,632 file.None.0xffffffff8000f886e0.dat
07/31/2026 10:12 AM 262,144 file.None.0xffffffff8001086b20.vacb
07/31/2026 10:12 AM 70,656 file.None.0xffffffff80021a1600.img
                3 File(s) 402,432 bytes
                2 Dir(s) 22,537,543,680 bytes free

C:\Users\hp\OneDrive\Documents\Task\DFIR\MemoryForensics\MemLabs-Lab5>

```

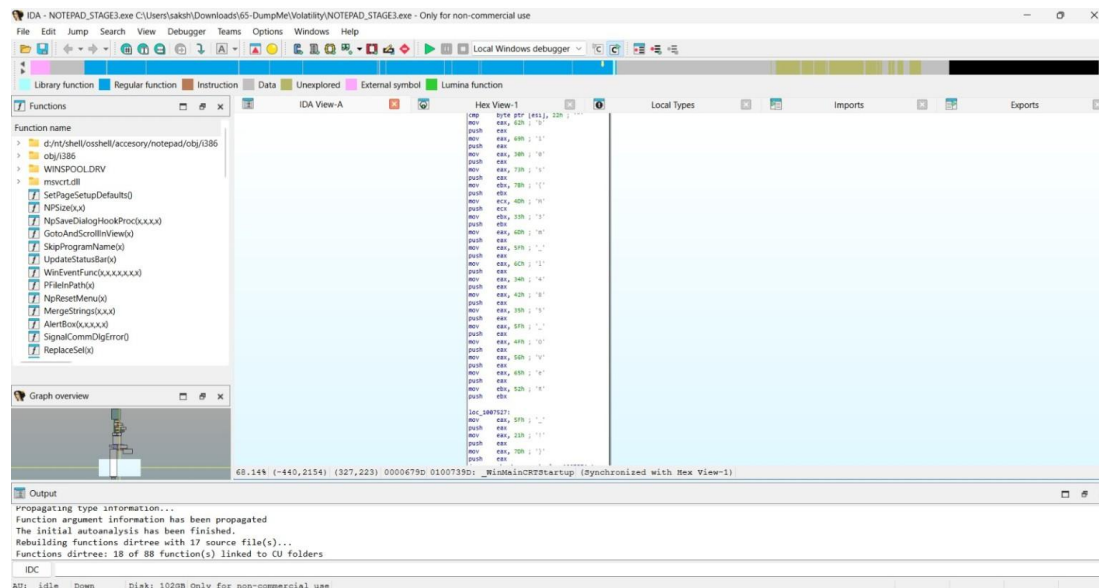
Step 21: Rebuild the extracted image as a runnable executable

```
copy notepad-dump\file.None.0xffffffff80021a1600.img NOTEPAD_STAGE3.exe
```

Why used: The .img file dumped above is the ImageSectionObject of NOTEPAD.EXE as it was mapped into memory, which is effectively the executable's own PE image. Copying/renaming it with an .exe extension reconstructs it into a file that can be loaded into a disassembler for static analysis, since the original notepad.exe on disk had been modified/replaced as part of the challenge.

Step 22: Reverse-engineer the executable to extract the final flag (IDA Pro)

Why used: NOTEPAD_STAGE3.exe was opened in IDA Pro and disassembled to look for logic added to the legitimate Notepad code. Inside WinMainCRTStartup, a long sequence of mov eax/ebx/ecx, <byte>; 'char' followed by push instructions was found — a classic stack-string obfuscation technique used to hide a string (here, the final flag) from static string-scanning tools, since the characters only exist in memory as immediate values inside instructions rather than as a plain ASCII string in the file.



Reading the single-byte immediate values in the order they are pushed reconstructs the final flag of the challenge:

```
bi0s{M3m_l4B5_0VeR_!}
```

4. Why Volatility 2 Was Needed After Volatility 3

Volatility 3 is the actively maintained, modern rewrite of the framework and was used first because it auto-detects the OS profile and is generally faster for initial triage (pslist, pstree, cmdline, filescan, dumpfiles, vadinfo, dlllist, netscan). However, Volatility 3 has not yet re-implemented every plugin from Volatility 2 — in particular consoles, cmdscan, and iehistory, which recover typed command history and browser history, were still only available in Volatility 2 (2.6) at the time of this investigation. Volatility 2's dumpfiles also supports the -Q/-D image-section dump used in Stage 3, which was needed to reconstruct a loadable copy of NOTEPAD.EXE for reverse engineering. Since the remaining clues in this challenge were held in typed console commands and in the executable's own code rather than in plain files or process metadata, the investigation was switched to Volatility 2.6 (with the profile identified via imageinfo as Win7SP1x64) to reach the final flag.

5. Conclusion

By combining Volatility 3 for fast initial triage with Volatility 2.6 for its legacy console-recovery and image-section-dump plugins, plus static reverse engineering in IDA Pro, the memory dump MemoryDump_Lab5.raw was fully analysed across three stages. Suspicious processes, a hidden RAR archive, a Base64-encoded string, and an obfuscated stack string inside the reconstructed NOTEPAD.EXE were all located and decoded, yielding three flags in total:

Stage 1: `flag{!!_w3LL_d0n3_St4g3-1_0f_L4B_5_D0n3_!!}`

Stage 2: `flag{W1th_th1s_$taGe_2_1s_c0mPL3T3_!!}`

Stage 3 (Final): `bi0s{M3m_l4B5_0VeR_!}`

This confirms the value of combining Volatility 3, Volatility 2, and static reverse-engineering tools in a memory forensics investigation: Volatility 3 for speed and modern plugin support, Volatility 2 for the legacy plugins and image-section recovery it has not yet replicated, and a disassembler for artefacts that were hidden in code rather than in memory-resident files.